

# Mixed Computing Trap Special

fraction + decimal Cross-Topic · 65 minutes · 1-to-3 Online Lesson

**Corresponding textbooks:** "New Thinking in Primary School Mathematics (Second Edition)" 5th Volume A Unit 1 + Modern Education 5th Class A Unit 1-3

**Core Traps:** □ T2 Decimal Operations · T9 Fractional Operations - Non-uniform format direct calculation · Decimal point misalignment

**SSPA Association:** ● **Extremely High Frequency** Approximately 25% of the scored test paper contains a mixture of fractions + decimals, which is the most likely unit to lose points in the entire paper

**Prerequisite knowledge:** Chapter 7-10 (four principles of fractions) · Chapter 12-13 (multiplication and division of decimals) · Chapter 16 (comprehensive fractions and decimal equations)

**Objectives of this class:** ❶ Unified format strategy (convert all fractions vs. all decimals) ❷ Mix the order of four arithmetic operations ❸ Cross-topic application questions

Student Name:

---

Class:

---

Date:

---

Time Spent:

---

## I.Warm-Up Questions (total 5 question , 5 minutes)

| # | Question                                                                                                  | Difficulty | Working Space (Show full working) |
|---|-----------------------------------------------------------------------------------------------------------|------------|-----------------------------------|
| 1 | Convert 0.5 into a fraction (simplest form)                                                               | Basic      |                                   |
| 2 | Convert $\frac{3}{4}$ into decimal                                                                        | Basic      |                                   |
| 3 | Calculate $0.25 + \frac{1}{2} = ?$ (Tip: unify the format first)                                          | Advanced   |                                   |
| 4 | Calculate $\frac{3}{5} \times 0.2 = ?$ (Tip: Multiply in a unified format)                                | Advanced   |                                   |
| 5 | Judgment: $\frac{1}{3} + 0.5$ Should the fraction be converted first or the decimal converted first? Why? | Advanced   |                                   |

## II.Core Knowledge + Worked Examples

### Knowledge Point 1: Mixed Addition of Fractions and Decimals — The strategy of unifying the format first ● SSPA

- ① **Core Principle** |||SEP|||: **Fractions and decimals cannot be added directly! The format must be unified first**  
 Strategy 1 (converting fraction method) |||SEP|||: Convert decimals into fractions → common fraction → calculation → reduction. Applicable to: when decimals can be converted into simple fractions (0.5, 0.25, 0.75, 0.2, 0.125, etc.)
- ② **Strategy 2 (conversion to decimals method)** |||SEP|||: **Convert fractions into decimals → Align decimal points → Calculate.** Applicable to: **When fractions can be converted into finite decimals (the denominator is 2,4,5,8,10,20,25...)** The key to choosing a strategy |||SEP|||: See which conversion result is more concise. If the denominator is 3,6,7,9, it can usually be converted to a fraction; if the denominator is 2,4,5,8, it can be converted to a decimal.
- ③ **Strategy 2 (conversion to decimals)**: Convert fractions into decimals → Align decimal points → Calculate. Applicable to: when the fraction can be converted into a finite decimal (the denominator is 2,4,5,8,10,20,25...)
- ④ **The key to choosing a strategy**: See which conversion result is more concise. If the denominator is 3,6,7,9, it can usually be converted to a fraction; if the denominator is 2,4,5,8, it can be converted to a decimal.

### □ Example of trap detonation (the most important demonstration in this class)

calculate:  $\frac{1}{4} + 0.5 = ?$

#### ✗ Common mistakes (60% students)

$0.25 + 0.5 = 0.3$  (the decimal point is misplaced)

Directly correct the decimal places of 0.25 and 0.5, and miscalculate as  $0.25 + 0.05 = 0.3$

#### ✓ Correct solution (conversion to fraction method)

$$0.5 = \frac{1}{2} \rightarrow \frac{1}{4} + \frac{1}{2} = \frac{1}{4} + \frac{2}{4} = \frac{3}{4}$$

 **Tip: "If the format is not consistent, unify it first. The denominator 3/6/7/9 can be converted into a fraction. The denominator 2/4/5/8 can be a decimal. The answer should be the simplest."**

 **The most frequent error: adding fractions and decimals directly without conversion, adding the decimal to the numerator as the answer**

 **The second most frequent error: Decimal point alignment error - tenth place should be aligned with tenth place, and hundredth place should be aligned with hundredth place**

### Knowledge Point - Worked Examples (write out unified format → calculateprocess)

| #         | Question                 | Difficulty                                                                        | Working Space |
|-----------|--------------------------|-----------------------------------------------------------------------------------|---------------|
| Example 1 | $0.75 + \frac{1}{4} = ?$ |  |               |
| Example 2 | $\frac{3}{5} + 0.2 = ?$  |  |               |

### Knowledge point 2: Mixed multiplication and division of fractions and decimals SSPA required test

#### Mixed multiplication and division strategies:

- ① **Recommended strategy: Convert all fractions**- After converting the fractions to decimals, use fraction multiplication (numerator times numerator, denominator times denominator), and finally reduce
  - ② Division: After converting fractions " $\div$  fraction =  $\times$  reciprocal"
- If converting to a decimal: Pay attention to the rules for the number of decimal places in decimal multiplication (a digit  $\times$  b digit = a+b decimal place)
- ④ **Key judgment |||SEP|||: If the decimal is a recurring decimal (such as 0.333...), it must be converted to a fraction.:** If the decimal is a recurring decimal (such as 0.333...), it must be converted to a fraction.

#### Example questions (trap demonstration)

Example 3:  $\frac{2}{3} \times 0.5 = ?$

#### example

Example 4:  $0.75 \div \frac{3}{4} = ?$

### Knowledge Point 2 Synchronization practice

| # | Question | Difficulty | Working Space |
|---|----------|------------|---------------|
|---|----------|------------|---------------|

|    |                                |                                                                                   |  |
|----|--------------------------------|-----------------------------------------------------------------------------------|--|
| 6  | $\frac{1}{4} \times 0.8 = ?$   |    |  |
| 7  | $0.6 \times \frac{5}{6} = ?$   |  |  |
| 8  | $\frac{3}{8} \div 0.25 = ?$    |  |  |
| 9  | $0.4 \div \frac{2}{5} = ?$     |  |  |
| 10 | $\frac{1}{2} \times 0.125 = ?$ |  |  |

### Knowledge point 3: Fractions + decimals in four mixtures (first multiplication and division, then addition and subtraction + unified format) SSPA Advanced

Mix four strategies (double challenge):

- ① **Step 1: Follow the order of operations**—Multiplication and division first, then addition and subtraction, brackets are calculated first if there are parentheses
- ② **Step 2: Unify the format of segments**—Convert multiplication and division parts to fractions uniformly, and decide to convert addition and subtraction parts to fractions or decimals
- ③ **Recommended overall strategy:** It is safest to convert all fractions (to avoid the problem of endless decimal divisions)
- ④ If there are parentheses and the parentheses contain fractions + decimals: first process the format within the parentheses to be unified

#### example

Example 5:  $0.5 + \frac{1}{3} \times 0.6 = ?$

#### example

Example 6:  $(\frac{1}{2} + 0.25) \times \frac{4}{5} = ?$

### Knowledge Point 3 Synchronization practice

| #  | Question                             | Difficulty                                                                          | Working Space |
|----|--------------------------------------|-------------------------------------------------------------------------------------|---------------|
| 11 | $0.3 + \frac{1}{2} \times 0.4 = ?$   |  |               |
| 12 | $\frac{2}{3} \times 0.75 + 0.25 = ?$ |  |               |

### III. Lesson Layered Synchronization Practice

Basic layer (total 5 questions, all must do - per question mixed fraction and decimal)

| #  | Question                 | Difficulty                                                                        | Working Space |
|----|--------------------------|-----------------------------------------------------------------------------------|---------------|
| 13 | $0.5 + \frac{1}{3} = ?$  |  |               |
| 14 | $\frac{1}{2} + 0.25 = ?$ |  |               |
| 15 | $0.75 + \frac{1}{8} = ?$ |  |               |
| 16 | $\frac{3}{10} + 0.4 = ?$ |  |               |
| 17 | $0.2 + \frac{2}{5} = ?$  |  |               |

Advanced layer (total 7 questions,   choose do - mix multiplydivide + four)

| #  | Question                             | Difficulty                                                                          | Working Space |
|----|--------------------------------------|-------------------------------------------------------------------------------------|---------------|
| 18 | $\frac{1}{4} \times 0.6 = ?$         |  |               |
| 19 | $0.8 \times \frac{3}{4} = ?$         |  |               |
| 20 | $\frac{5}{6} \div 0.5 = ?$           |  |               |
| 21 | $0.3 + \frac{1}{4} \times 0.8 = ?$   |  |               |
| 22 | $\frac{1}{2} \times 0.6 + 0.25 = ?$  |  |               |
| 23 | $1.25 - \frac{1}{4} \div 0.5 = ?$    |  |               |
| 24 | $(0.5 + \frac{1}{3}) \times 0.6 = ?$ |  |               |

 challenge layer (total 5 questions,   choose do—more step mixing + trap)

| #  | Question                                             | Difficulty                                                                            | Working Space |
|----|------------------------------------------------------|---------------------------------------------------------------------------------------|---------------|
| 25 | $0.75 \times \frac{2}{3} + \frac{1}{6} = ?$          |  |               |
| 26 | $\frac{3}{4} \div 0.25 + 0.5 \times \frac{1}{3} = ?$ |  |               |

|    |                                                     |                                                                                     |  |
|----|-----------------------------------------------------|-------------------------------------------------------------------------------------|--|
| 27 | $1.2 \times (\frac{5}{6} - 0.25) = ?$               |    |  |
| 28 | $(0.6 + \frac{1}{5}) \div (\frac{1}{2} - 0.1) = ?$  |  |  |
| 29 | $0.125 + \frac{3}{8} \times 0.4 - \frac{1}{10} = ?$ |  |  |

**Knowledge point 4: Cross-topic application questions (shopping discounts + recipe portions + engineering measurements)  SSPA compulsory exam**

**Four steps to solve the problem:**① **Circle keywords**("Total" "After discount" "Each portion") ②**Column**(Uniform format for fractions and decimals) ③**Calculation**(Pay attention to unit conversion) ④**Write a complete answer**(Step points will be deducted if you don't write an answer!)

**example**

**Example 7:** A cake originally priced at \$80 is now discounted by 20% (i.e.  $\frac{4}{5}$ ). Xiao Ming has 0.6 pieces of cake. How much does he need to pay?

applicationquestionpractice (all must do, column → unified format → calculate → answer sentence)

| #  | Question                                                                                                                                  | Difficulty                                                                            | Working Space |
|----|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------|
| 30 | A pencil $\frac{1}{2}$ yuan, Xiao Ming bought 0.6 pieces (billed on a pro-rata basis). How much does he need to pay? (Answer in decimals) |  |               |
| 31 | Recipe requires $\frac{3}{4}$ kg flour and 0.25 kg sugar. How many kilograms do the two materials weigh in total?                         |  |               |
| 32 | A ribbon is 2.5 meters long and uses $\frac{1}{4}$ to make a bow. How many meters of ribbon were used for the bow?                        |  |               |

advancedapplicationquestion (  choose do, SSPAPaper 2commonquestion type)

| #  | Question                                                                                                                                                                      | Difficulty                                                                            | Working Space |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------|
| 33 | A jacket, originally priced at \$360, is 25% off (0.75). In addition, use membership $\frac{1}{10}$ coupons to get discounts. How much does Xiao Ming have to pay in the end? |  |               |

|    |                                                                                                                                                                                                                                                                                                            |                                                                                     |  |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--|
| 34 | Recipe: Cake requires $\frac{2}{3}$ cups of milk. How many cups of milk are needed if only 0.75 servings (i.e. $\frac{3}{4}$ servings) are made?                                                                                                                                                           |    |  |
| 35 | The engineering team completed the project in one day    SEP   . After 0.5 days, another 0.125 of the project was completed. What fraction of the project has been completed? $\frac{1}{8}$ . After 0.5 days, another 0.125 of the project was completed. What fraction of the project has been completed? |  |  |
| 36 | The water bottle originally contained 0.75 liters of water. After drinking $\frac{1}{3}$ liters, an additional 0.2 liters was added. How many liters does the water bottle have now?                                                                                                                       |  |  |

## IV. Ultimate challenge area

| #                                                                                        | Question                                                                                                                                                                                                                                                                                                                                                                                                                                     | Difficulty                                                                            | Working Space |
|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------|
| <br>1   | $(0.75 + \frac{1}{6}) \div (\frac{2}{3} - 0.25) \times 1.2 = ?$                                                                                                                                                                                                                                                                                                                                                                              |    |               |
| <br>2   | A water tank has a capacity of 5 liters. The first injection of $\frac{1}{4}$ is 0.8 of the cylinder and the second injection is 1.25 liters. How many liters are there in the cylinder now?                                                                                                                                                                                                                                                 |    |               |
| <br>3 | Three stores: store A' s discount is    SEP   , store B' s discount is 0.65, and store C' s discount is    SEP   . Which one is the cheapest? Please prove it with calculations. (Tip: Convert all to decimals for comparison) $\frac{3}{5}$ , the discount for store B is 0.65, and the discount for store C is $\frac{2}{3}$ . Which one is the cheapest? Please prove it with calculations. (Tip: Convert all to decimals for comparison) |  |               |

## V. Class afterhomework

Basic must doquestion (total 6 questions, perquestionmust mix fractionanddecimal)

| #  | Question                     | Difficulty                                                                            | Working Space |
|----|------------------------------|---------------------------------------------------------------------------------------|---------------|
| H1 | $0.5 + \frac{1}{4} = ?$      |  |               |
| H2 | $\frac{2}{5} + 0.3 = ?$      |  |               |
| H3 | $\frac{3}{4} \times 0.4 = ?$ |  |               |
| H4 | $0.6 \div \frac{3}{5} = ?$   |  |               |

|    |                                                                                                                                                |                                                                                     |  |
|----|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--|
| H5 | $0.3 + \frac{1}{2} \times 0.4 = ?$                                                                                                             |    |  |
| H6 | A kettle holds 1.5 liters of water and uses    SEP   . How many liters of water were used? $\frac{1}{3}$ . How many liters of water were used? |  |  |

Advanced choose doquestion (total 3 questions,  choose do)

| #  | Question                                                                                                                                                        | Difficulty                                                                          | Working Space |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------|
| H7 | $(0.5 + \frac{1}{6}) \times \frac{3}{4} = ?$                                                                                                                    |  |               |
| H8 | Compare $\frac{2}{3} \times 0.75$ and $0.6 \times    SEP   $ , which is larger? How much difference? $\frac{5}{6}$<br>Which one is larger? How much difference? |  |               |
| H9 | An item costs \$240, first add $\frac{1}{4}$ (i.e. $\times 1.25$ ), and then get a 15% discount (0.85).<br>What is the final selling price?                     |  |               |

## VI. The Lessoncorecommon errorssummary

 **Review of Learning Objectives - After completing this lesson you should be able to:**

☐ Identify all trap types in our hall

☐ Solve  basic questions independently (100% correct)

☐ Challenge  Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

| # | common error                                                                                                           | Correct Approach                                                                                                |
|---|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 1 | <b>Fractions and decimals are added directly without conversion</b> : $0.5 + \frac{1}{3} = 1.5/3$ (random combination) | First unify the format: convert all fractions or all decimals, and then calculate                               |
| 2 | <b>Decimal point alignment error</b> : $0.25 + 0.5 = 0.3$                                                              | Tenth place to tenth place, percentile to percentile, add 0 if necessary                                        |
| 3 | <b>When converting a fraction into a decimal, the divisions cannot be divided but the approximation is taken</b>       | The denominator contains 3, 6, 7, 9, etc. → Convert to fraction, do not convert to decimal                      |
| 4 | <b>Mixing four ignores "multiplication and division first, then addition and subtraction"</b>                          | Circle the multiplication and division parts first, and then add and subtract after completing the calculation. |
| 5 | <b>Wrong number of decimal places in decimal multiplication</b>                                                        | a place                                                                                                         |
| 6 | <b>Fraction division without reciprocal conversion</b>                                                                 | $\div$ Fraction = $\times$ The reciprocal of the fraction (swapping the numerator and denominator)              |
| 7 | <b>The answer to the application question does not include the unit or the answer is incomplete.</b>                   | Must write "Answer:..." + unit, otherwise step points will be deducted                                          |